5. The variance of random variable X is 4 its mean is 2 then $E(X^2)$ is

(a) 8 (b)6 (c) 4 (d) 2

6. Var $(4X+3)$ is

(a) 7 Var (X) (b) 16 Var (X) (c) 256 Var X (d) 0

7. The second order moment about mean is

(a) Mean (b)standard deviation (c)Variance (d) median

8. If variance = 20, second order moment about the origin = 276, then the mean of the random variable X is

(a) 16 (b)5 (c)4 (d) 2

II. Fill in the Blanks:

9. A variable whose values are real numbers and determined by the outcome of the random experiment is called _____

10. $F(X)$ is a distribution function then $F(-\infty)$ _____

11. Let X be a random variable and for any real constant a and b and $a \leq b$ then

$P(a \leq X \leq b) =$ ____

12. In the case of continuous random variable the probability at a point is always _____

13. If $F(X)$ is a cumulative distribution function of a continuous random variable X then $F'(X) =$ _____

14. $f(x)$ is a p.d.f. of continuous random variable then $\int_{-\infty}^{\infty} xf(x)\,dx$ ___

15. The Mathematical Expectation of a random variable X is also called as _____

III. Answer shortly:

16. Distinguish between discrete random variable and continuous random variable

17. Distinguish between probability mass function and probability density function.

18. Define mathematical Expectation of a discrete random variable.